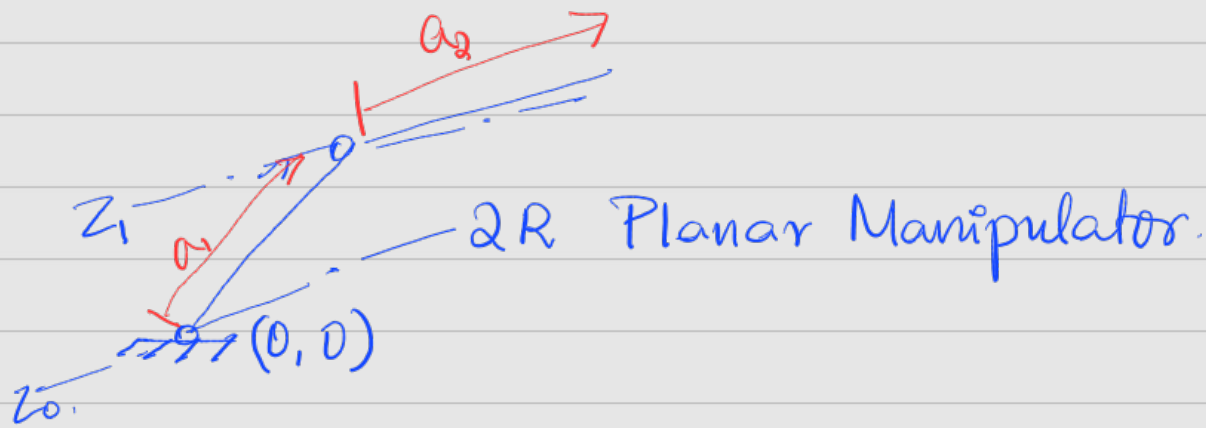


2 DOF FK using DH Parameters.



Apply DH convention.

z_i $\rightarrow$ Axis of rotation for link $i+1$.

x_i $\rightarrow$ lies along links.

a_i $\rightarrow$ link length

α_i $\rightarrow$ link twist.

Build DH table.

Link	a_i (link length)	α_i (link twist)	d_i (link off)	θ_i
1	a_1	0	0	θ_1
2	a_2	0	0	θ_2

Given transformation matrix A_i

$$A_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & a_1 c\theta_1 \\ s\theta_1 & c\theta_1 & 0 & a_1 s\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & a_2 c\theta_2 \\ s\theta_2 & c\theta_2 & 0 & a_2 s\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2^0 = A_1 \times A_2$$

$$= \begin{bmatrix} c(\theta_1 + \theta_2) & -s(\theta_1 + \theta_2) & 0 & a_1 c\theta_1 + a_2 c(\theta_1 + \theta_2) \\ s(\theta_1 + \theta_2) & c(\theta_1 + \theta_2) & 0 & a_1 s\theta_1 + a_2 s(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$x = a_1 c\theta_1 + a_2 c(\theta_1 + \theta_2)$$

$$y = a_1 s\theta_1 + a_2 s(\theta_1 + \theta_2)$$

$$z = 0$$

$$\text{Orientation } \phi = \theta_1 + \theta_2$$

